

Kunj Pandya

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MS Computer Science · graduating May 2027 · Hoboken, NJ · open to relocation

EDUCATION

Stevens Institute of Technology

Master of Science in Computer Science

Hoboken, NJ

Sept 2025 – May 2027

- **Coursework:** Distributed Systems, Operating Systems, Concurrent Programming, Parallel Computing (CUDA), Machine Learning, Database Management Systems, Cloud Computing

D.J. Sanghvi College of Engineering, Mumbai University

Bachelor of Technology in Artificial Intelligence and Machine Learning

Mumbai, India

Sept 2021 – May 2025

- **Coursework:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures & Algorithms

EXPERIENCE

Software Engineer Intern, iOS (AI Features)

Fini

May 2026 – Present

Remote

- Build **AI-assistant features** in **Swift and SwiftUI** (MVVM, Swift Package Manager) that turn natural-language intent into **structured actions** for task assignment and routine building.
- Migrate legacy **UIKit** → **SwiftUI** screens behind **feature flags** in a production codebase with **~15 active contributors**.
- Ship through **senior-reviewed PRs** with atomic commits, rebase-before-merge, SwiftLint-enforced style, and constants-driven localization across **English, German, and Spanish**.

Graduate Research Assistant, Distributed Systems

Stevens Institute of Technology

Nov 2025 – Present

Hoboken, NJ

- Research **energy efficiency in heterogeneous federated learning** (HeteroFL), isolating where cold-start training wastes energy.
- Built a **distributed profiling pipeline** (**TensorFlow.js**, **Socket.IO**, **Redis**) that collects real-time **per-layer power metrics** from **100+ heterogeneous edge devices**.
- Traced a **40% cold-start energy overhead** to bottleneck layers drawing **3×** higher consumption; the resulting optimizations cut training energy **35% at <2% accuracy loss** (submitted for publication).

Research Intern – Placement Intelligence Platform

D.J. Sanghvi College of Engineering

May 2024 – June 2025

Mumbai, India

- Owned the **Node.js backend** for a placement platform serving **500+ students**, designing REST APIs that handled **10K+ requests/day at <200 ms p95** under concurrent resume-upload workloads.
- Trained a custom **spaCy NER** resume parser on **2K labeled resumes** (70/15/15 split) by transfer learning from **en_core_web_lg**, reaching **95% F1** against a **78% regex baseline**; served it through **FastAPI**.
- Fine-tuned **MobileNetV2** on a custom behavioral-engagement dataset for real-time interview-confidence detection at **30 FPS and 88% accuracy**, cutting inference latency **40%** with quantization.

Undergraduate Research Assistant

D.J. Sanghvi College of Engineering

June 2022 – May 2024

Mumbai, India

- Co-authored a **peer-reviewed paper** in *Library Progress International* (2024) quantifying a **30% reduction** in simulated household food waste; **first-authored its system architecture section**.
- Trained the **TensorFlow** ingredient-recognition model on **10K+ annotated food images**, later reimplemented as NutriSense.

PROJECTS — DISTRIBUTED SYSTEMS & BACKEND

NexusMesh — Decentralized P2P Event Mesh | Java 17, gRPC, Protocol Buffers, Maven

2025 – 2026

- Implemented a **Chord DHT from primitives** — **160-bit SHA-1** ring, finger-table routing, **$O(\log^2 N)$** stabilization — keeping lookups correct through node joins, failures, and key migration with **no central broker**.
- Layered a pub/sub event broker over **bidirectional gRPC streaming**; benchmarked **<5 ms p99 lookup across a 100-node network**, **<1 ms** local event delivery, and **±5%** storage-distribution variance.

TaskPulse — Distributed Task Orchestrator | Python, FastAPI, Redis, PostgreSQL, Docker

2025 – 2026

- Built a horizontally scalable orchestrator (job submission, priority scheduling, status tracking) over a **Redis priority queue** with PostgreSQL as durable state, sustaining **10K+ tasks/day at 99.9% reliability**.
- Resolved **task starvation and retry storms** under burst load with **priority-aware batching**, **idempotent execution**, and **bounded exponential backoff**.

VaultChat — Zero-Knowledge Encrypted Messaging | Go, Redis Pub/Sub, PostgreSQL, Web Crypto API

2026

- Built **stateless Go WebSocket nodes** (goroutines, gorilla/websocket) that fan messages out over **Redis Pub/Sub**, subscribing to a room channel only when a local client is present.
- Enforced a **zero-knowledge server** with **ECDH P-256** key exchange and **AES-GCM-256** (non-extractable keys, 12-byte nonces); PostgreSQL stores ciphertext only, with **no plaintext field** in schema or wire protocol.

PROJECTS — AI, ML & APPLIED SYSTEMS

Edara — Procedure Intelligence Engine <i>Python, FastAPI, Supabase, Redis, OpenAI</i>	2026 – Present
- Built a deterministic staged pipeline (s0_ingest → s9_project) that turns technical PDFs into structured, source-grounded task data, with versioned artifact contracts and validation at every stage boundary. - Confined the LLM to semantics with mandatory span citation; a provenance resolver maps each claim to its source span, and unresolvable provenance fails the build (13 test modules, golden ground truth).	
DocLens — Edge-Native Doc Search & Chat <i>TypeScript, Cloudflare Workers, Vectorize, Gemini</i>	2026
- Shipped a live RAG service on Cloudflare’s edge: BGE embeddings in Vectorize, streaming Gemini 2.0 Flash answers, and Durable Objects holding chat sessions on SQLite. - Measured ~70 ms cold start and 200–400 ms end-to-end search, and cut the client bundle from 1.5 MB to 285 KB gzipped.	
FleetIQ — Vehicle Entry/Exit Matching <i>Python, YOLOv8, OpenCV, EasyOCR, SQLite</i>	2025
- Built a four-stage pipeline at HackHouse SF (pitched to AirGarage): metadata parsing, YOLOv8 plate detection, multi-engine OCR, and fuzzy temporal matching on Levenshtein distance within a 72-hour window. - Reached 98.15% OCR success (1,960/1,997 images) and 99.85% bounding-box extraction at ~115 images/sec with six enhancement strategies, producing 591 matched pairs.	
TradeFlow — Paper Trading Platform <i>TypeScript, React, Node.js, PostgreSQL, TensorFlow.js</i>	2026
- Built a full-stack trading simulator with real-time Alpaca pricing across the S&P 500, limit/stop orders on 60-second validation cycles, FIFO cost-basis tax tracking, and group competitions. - Added a TensorFlow.js ensemble (Holt-Winters, LSTM, GRU) over 23 engineered OHLCV features and an SSE-streamed LLM research pipeline (DeepSeek-R1 via Ollama/Groq, Gemini).	
NutriSense — Ingredient Recognition & Recipes <i>Python, TensorFlow, MobileNetV2, Streamlit</i>	2024
Implemented the published system: MobileNetV2 at 96.41% test accuracy across 36 ingredient classes, and a hybrid collaborative + TF-IDF recommender over 8,269 recipes at 93% precision / 89% recall.	
RecallAI — Semantic Memory for YouTube <i>JavaScript, transformers.js, Chrome MV3, IndexedDB</i>	2026
Built a Chrome extension that captures live captions and computes local all-MiniLM-L6-v2 embeddings, linking related videos in a 3D memory graph with no server or API keys.	
Credit Risk Inference System <i>Python, XGBoost, scikit-learn, Flask, pytest</i>	2026
Trained an XGBoost pipeline on a 1,000-row synthetic credit dataset to 0.91 AUC-ROC and 81% accuracy, served through a Flask REST API at <200 ms median latency and verified with a 1,000-request stress test.	

PROJECTS — PRODUCT, WEB & GAMES

I Wizard — 1st Place, QuackHacks '26 <i>real-time 3D dueling game in React Three Fiber, built in 24 hours</i>	2026
Wizard Duel — Unity 6, C# <i>10-level campaign, 8 spells, voice-controlled casting via Whisper, state-machine AI</i>	2026
QuickFill — Chrome Extension (MV3) <i>snippet and autofill extension on the Chrome Web Store, 19+ active users</i>	2026

TECHNICAL SKILLS

Languages: Python, Java, Go, C/C++, TypeScript/JavaScript, Swift, SQL, Bash
Distributed Systems & Backend: Chord DHT, gRPC/Protobuf, REST APIs, FastAPI, Node.js/Express, Redis (queues, pub/sub, caching), PostgreSQL, WebSockets/Socket.IO, idempotency, backoff, concurrency
ML & AI: PyTorch, TensorFlow/TensorFlow.js, scikit-learn, XGBoost, spaCy NER, YOLOv8, MobileNetV2, RAG & vector search, LLM integration, federated learning, CUDA (coursework)
Cloud & Infrastructure: Docker/Compose, AWS (EC2, S3), Cloudflare Workers, Durable Objects, Workers AI, Supabase, Vercel, GitHub Actions, Linux
Mobile & Frontend: SwiftUI, UIKit, React, Next.js, Tailwind CSS, React Three Fiber, Unity (C#)
Engineering Practice: Git & PR review, feature flags, pytest, JUnit, Maven, profiling & benchmarking, Agile

PUBLICATIONS

S. Shah, D. Chande, K. Pandya , K. Gopani, A. Gawade, N. Rathod, A. Florence. “Revolutionizing Culinary Experiences: AI-Driven Ingredient Recognition and Personalized Recipe Recommendation System.” <i>Library Progress International</i> , Vol. 44, No. 3, pp. 14279–14290, 2024.
Energy efficiency in heterogeneous federated learning (cold-start overhead). Submitted for publication, 2026.

AWARDS & CERTIFICATIONS

1st Place, QuackHacks '26 <i>24-hour hackathon, won with I Wizard</i>	Feb 2026
Top 100 Builders, HackHouse SF <i>resident; built FleetIQ and pitched it to AirGarage</i>	Dec 2025
OCI 2025 Certified AI Foundations Associate <i>Oracle Cloud Infrastructure</i>	2025
Also completed: Java Essential Training: Objects and APIs · Claude Code in Action · Hands-On Introduction: Python	

LEADERSHIP, COMMUNITY & ADDITIONAL EXPERIENCE

Technical Lead, ACM Student Chapter <i>D.J. Sanghvi College of Engineering, Mumbai</i>	Dec 2023 – Jan 2025
Lead Member, Prayaas Foundation <i>led ~15 drives a year (education, food, health) with teams of 15–35</i>	2021 – 2024
Public Relations, Gujarati Intellectual Movement <i>secured sponsors for lectures, festivals, and drama</i>	2022 – 2024
Agent Assistant, Pandya Estate <i>client enquiries, site visits for 3–4 clients a week, document verification</i>	2021 – 2025